# Type Testing of Process Valve Packing for Fugitive Emissions

API STANDARD 622
SECOND EDITION, AUGUST 2011

DRAFT

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**Downstream Segment**

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## INTRODUCTION

The purpose of this API Standard is to establish a uniform procedure for evaluation of process valve stem packing. The testing approaches defined within this standard provide a method for evaluating packing systems. This testing program will provide a basis for the comparison of the emissions and life cycle performance of packing.

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# Type Testing of Process Valve Packing for Fugitive Emissions

## 1 Scope

This API Standard specifies the requirements for comparative testing of valve stem packing for process applications where fugitive emissions are a consideration. High temperature packing(s) shall be suitable for use at service temperatures –29°C to 538°C (–20 °F to 1000 °F) and temperature limited packing(s), suitable for manufacturer specified temperature range. Factors affecting fugitive emissions performance that are considered by this Standard include temperature, pressure, thermal cycling, mechanical cycling and corrosion.

This standard is not intended to replace type testing of valve assemblies or valve production testing.

**1.1** This standard establishes requirements and parameters for the following tests.

a. Fugitive emissions

b. Corrosion

c. Packing material composition and properties

**1.2** Test methods apply to valve packing for use in on-off valves with the following stem motion(s):

a. Rising stem

b. Rotating stem

c. Rising rotating stem

**1.3** The test for fugitive emissions is based upon elements of EPA Method 21, providing comparative values of packing performance.

**1.4** Packing tested to this standard may or may not perform the same when used in a manufacturer's valve. Performance to a specified emission limit may be confirmed by re-testing the packing in an actual valve in accordance with Section 4.4 of this Standard.

## 2 Reference Publications

The following standards contain provisions that, through reference in this text, constitute provisions of this API Standard. At the time of the publication, the editions indicated were valid. All standards are subject to revision, and parties to agreements based on this Standard are encouraged to investigate the possibility of applying the most recent editions of the standards indicated below.

Applicable reference standards: API Standards 599, 600, 602, 603, 608, 609, 621, ASME B16.34 and MSS SP-96, MSS SP-120 and EPA Method 21.

## 3 Terms and Definitions

For the purposes of this standard, the following definitions shall apply.

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**3.1**
**active inhibitor**
A type of galvanic corrosion inhibitor typically applied as a surface coating to packing, providing a sacrificial anode material corroding in preference to the surrounding metal. See also “corrosion inhibitor, galvanic corrosion, passive inhibitor, pitting.”

**3.2**
**actuate**
To cause movement of a valve stem either by raising, lowering, or turning it.

**3.3**
**anti-extrusion ring**
A ring of packing used at one or both ends of a packing set to prevent extrusion of packing material into clearances.

**3.4**
**ambient temperature**
temperature that is below 40°C (110°F). The ambient temperature of each temperature cycle shall not vary by more than ±6°C (±10°F).

**3.5**
**axial**
In the direction of a shaft or stem axis.

**3.6**
**braided packing**
Packing typically constructed of intertwining strands of synthetic or natural fibers. Strands may consist of yarn or filaments, and may also include metallic materials. Two primary braid configurations are square and interbraided or interlocking.

**3.7**
**bolt torque**
The amount of twisting or turning effort (expressed as Nm, ft-lb or in-lb) required to turn the nuts on a gland flange, commonly used to describe the load that a gland flange exerts on a valve packing set.

**3.8**
**bore diameter**
The inside diameter into which packing is inserted; also called the stuffing box bore.

**3.9**
**bushing**
Cylindrical spacer used to take up excess space in a stuffing box.

**3.10**
**corrosion inhibitor**
An ingredient added to packing, which decreases the potential for galvanic corrosion in the stuffing box. Corrosion inhibitors may be classified as either passive or active. See also, “active inhibitor, galvanic corrosion, passive inhibitor, pitting.”

**3.11**
**die-formed packing**
A valve stem packing typically constructed from ribbons of graphite tape or braided packing that has been subjected to pre-compression with tooling of a specific geometry. This process changes the shape and density of the material from its original free shape and natural density to a defined shape and higher density.

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**3.12**
**EPA Method 21**
A leak check method established by the Federal Government Environmental Protection Agency (EPA) for performing emissions measurements on equipment such as valves, pumps, and flanges.

**3.13**
**eccentricity**
The distance that the central axis of a valve stem is offset from the central axis of the stuffing box through which it passes.

**3.14**
**emissions**
Gaseous or liquid leak given off by a piece of equipment. This is usually used in reference to volatile organic hydrocarbons (VOC's) and typically expressed in parts per million volumetric (ppmv, or ppm).

**3.15**
**galvanic corrosion**
An electro-chemical reaction that may occur between a metal and a material of a different chemical nobility, such as another metal, carbon, or graphite when both materials are exposed to an electrically conductive media. See also, "active inhibitor, corrosion inhibitor, passive inhibitor, pitting."

**3.16**
**gland**
A movable part which protrudes into a stuffing box to compress a packing set or packing ring.

**3.17**
**gland load**
The amount of load applied to a packing set.

**3.18**
**gland stud**
A threaded rod or eye-bolt, extending from a valve body against which the gland flange is tightened to compress a packing set.

**3.19**
**leak**
Measurable amount of test fluid escaping from the test gland.

**3.20**
**leak rate**
The quantity of test fluid passing through (or around) a seal in a given period of time.

**3.21**
**mechanical cycle**
A motion of the stem simulating the movement of a valve closure member from the fully closed position to the fully open position, and returning to the fully closed position. See also, "stroke."

**3.23**

**packing set**
A grouping of individual packing rings designed to fill the cavity of the valve stuffing box.

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**3.24**
**passive inhibitor**
A type of galvanic corrosion inhibitor blended into the raw graphite used in packing, providing a protective coating to prevent a galvanic reaction from occurring. See also, “active inhibitor, corrosion inhibitor, passive inhibitor, pitting.”

**3.25**
**pitting**
Surface cavities that occur on a metal as a result of galvanic corrosion or mechanical erosion. See also, “galvanic corrosion.”

**3.26**
**quarter-turn valve**
valve that will fully open or close with a nominal 90° rotation of the stem.

**3.27**
**rising stem valve**
valve in which the movement of the stem is in an axial direction, with no rotation.

**3.28**
**spacer bushing**
See “Bushing”

**3.29**
**stem**
metal rod that connects the internal closure member (, such as a disc or ball) of a valve to a handwheel, handle, or actuator.

**3.30**
**stroke**
One half of a mechanical cycle starting from either a fully open or fully closed position. See also, “mechanical cycle.”

**3.31**
**stuffing box**
space into which a compression packing is inserted. Also known as a packing chamber.

**3.32**
**surface finish**
measure of the roughness of a surface typically expressed in micro-inches or micro-meters.

**3.33**
**test valve**
valve used for verifying packing performance in a specific Manufacturer’s valve.

## 4 Fugitive Emissions Test

### 4.1 TEST FIXTURE

**4.1.1** The test stand packing gland is constructed as a fixture designed to simulate a typical valve. A test gland arrangement with a combined leak detection/gland flange is shown in Figure 1. The test fixture shall orient the stem in a horizontal position and shall be constructed so as not to affect test results. The use of bushings is allowed as necessary. Test stands may be equipped with multiple test fixtures. See Annex 1 for construction details.

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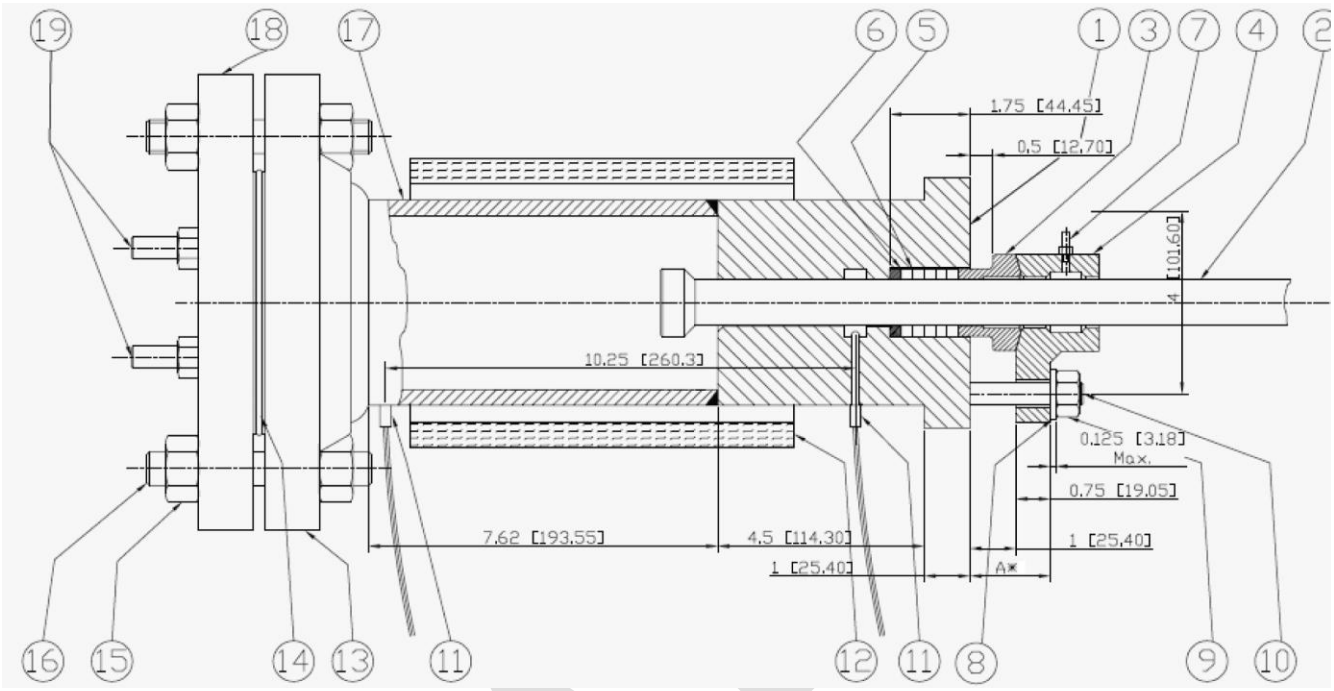



Figure 1

**4.1.1**

**4.1.1.1** Test fixture components

1) Bonnet
2) Stem
3) Gland
4) Gland flange
5) Stem packing
6) Bushing
7) Leak detection fitting
8) Washer
9) Gland nut
10) Gland stud
11) Thermocouple
12) Heating element

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13) Bonnet flange

14) Gasket

15) Flange stud nut

16) Bonnet studs

17) Housing

18) Blind flange

19) Gas inlet and outlet ports

**4.1.1.2** Test fixture packing gland dimensions and tolerances shall be as follows:

a) Stem diameter: 25.4 mm +0.0/–0.2 mm (1.000 in. +0.0/–0.008 in.)

b) Stem straightness: Max. 0.04 mm per 305 mm (0.0016 in. per 12 in.)

c) Stem cylindricity: 0.04 mm (0.0016 in.) Max.

d) Stem surface finish: 0.40 – 0.80 µm Ra (16-32 µ-in. Ra)

e) Stuffing box diameter: 38.1 +0.25/–0.0 mm (1.5 in. +0.010/–0.0 in.)

f) Stuffing box depth: 44.5 mm ±1.5748 mm (1.75 in. ±0.062 in.)

g) Stuffing box surface finish: 3.20 µm Ra +1.25/–0.625 µm (125 µ-in. Ra +50 /–25 µ-in.)

h) Gland bottom machined flat parallel to gasket surface: Max. 0.15 mm (0.006 in.)

i) Gland to stuffing box diametrical clearance: 0.13 – 0.38 mm (0.005 – 0.015 in.)

j) Stem to gland (flange) diametrical clearance: 0.5 – 0.8 mm (0.020 – 0.030 in.)

k) Gland stud diameter: 5/8 in. -11UNC (2 pieces, length per Figure 1)

**4.1.1.3** Test fixture components shall be constructed of carbon steel materials except for:

a) Stem: ASTM A182 Grade F6a Rc15-28.8 (200-275 HB)

b) Gland bolts/studs: ASTM A193 Grade B7

c) Gland nuts: ASTM A194 Grade 2H

d) Housing: DN 100 (4 in.) Schedule 80 Seamless Pipe or A-105

e) Flanges: DN 100 (4 in.) Class 300 Per ASME B16.5 (material group ASME B16.5 Table F2-1.1) or ASTM A105

**4.1.2** Mechanical Cycle:

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The test fixture shall be equipped with an actuator capable of stroking the test stem to simulate the mechanical cycle of a valve as follows:

a) Rising stem:

Rate: 3 mm to 5 mm (0.12 in. to 0.20 in.) per second

Stroke: 102 mm ±3 mm (4 in. ±0.12 in.)

b) Rotating stem:

Rate: 10° to 15° per second

Rotation: 90° ±5°

c) Rotating rising stem

Rate: 3 mm to 5 mm (0.12 in. to 0.20 in. ) per second

Stroke: 102 mm ±3 mm (4 in. ±0.12 in )

Number of rotations (to be confirmed with manufacturer based on 4" Class 300)

**4.1.3** External Loads

The actuator(s) shall not apply any transverse forces, i.e. side load to the test stem.

**4.1.4** Temperature Monitoring:

The test fixture(s) shall be equipped with thermocouples for continuously monitoring temperature during thermal cycling.

**4.1.5** Temperature shall be monitored and recorded at two locations.

a) At the flow line of the test chamber

b) Adjacent to the stuffing box

c) The thermocouple adjacent to the stuffing box shall control the test temperature between room temperature and maximum recommended temperature limit up to 260°C (500°F) as recommended by the packing manufacturer.

d) The temperature at the flow line thermocouple shall be the reference measurement.

**4.1.6** The fixture shall be heated using an external heat source, blanket, heating coils or other suitable equipment.

## 4.2 LEAK TEST EQUIPMENT SELECTION AND CALIBRATION

**4.2.1** Monitoring equipment shall be a flame ionization vapor analyzer capable of providing on-board data logging with digital readout. The equipment shall be certified as intrinsically safe for use with the test fluid.

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**4.2.2** The equipment shall meet the following performance requirements in the flame ionization mode using methane as the test fluid.

a) Variation: less than ± 2% at 100 ppm.

b) Dynamic range: 1.0 – 50,000 ppm.

c) Linear range: 1.0 – 10,000 ppm.

d) Minimum detectable level (defined as 2 x the peak noise): 300 ppb hexane

e) Maximum response time to reach final value: 3 seconds.

f) Maximum recovery time to return to 10% of initial value: 5 seconds

g) Sample flow rate at probe inlet: 1.14 – 1.51 l/min (0.30 – 0.40 gal/min)

**4.2.3** The testing equipment shall be calibrated according to the manufacturer's directions. A current record of test equipment calibration shall be maintained by the test facility.

**4.2.4** The test equipment shall be inspected prior to each use to insure against fouling of the detector probe. This shall be done to EPA Method 21, using an external calibration gas, with a known methane concentration.

### 4.3 PACKING SELECTION AND INSTALLATION

**4.3.1** Pre-qualification

Packing submitted for type testing shall be suitable for the conditions indicated in Section 1 "Scope" of this standard.

**4.3.2** Packing Selection

**4.3.2.1** Test packing shall be selected at random from either:

a) A standard production lot as supplied by the manufacturer, or

b) Distributor stock.

**4.3.3.1** Test packing shall be 6.3 mm (1/4 in.) cross-section for the test fixture. Validation of a random selection process shall be provided to the testing facility.~~(In the form of what?) Suggest removing this paragraph.~~ Packing in production shall be pulled from standard production line. Random selection not applicable for Prototype material.

**4.3.3** Packing Installation

A qualified laboratory representative or technician shall install the packing according to the manufacturer's standard installation instructions except that the maximum packing bolt stress shall not exceed 172,369 kPa (25,000 psi.), ~~this is applicable to packing test in fixture onlynot in a test valve~~ However, the packing stress (load) during the testing shall not exceed the packing manufacturer's recommended maximum value.

**4.3.3.1** For all packing installations for testing in packing test fixture.

a) All test fixture components shall be thoroughly cleaned with acetone or equivalent solvent prior to testing.

b) Any reconditioned parts shall be inspected and shall comply with the requirements of 4.1.2.

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c) Components shall be inspected for damage prior to assembly.

d) Caution shall be taken to avoid contact between the stem and gland.

e) Fasteners shall be lubricated.

f) Gland (flange) height measured from a specific datum shall be recorded.

g) Special preparation of the packing or assembly components is prohibited.

**4.3.3.2** For all packing installations when testing in test valves (intended for valve manufacturers only).

The test valve will either be prepared by the valve manufacturer or qualified laboratory representative. Packing installation procedures shall be provided by the packing manufacturer or valve manufacturer and shall include required packing gland bolt stress. A valve manufacturer can choose any size and class rating for test valves but should consider design differences that could impact packing behavior. For comparison purposes, two typical selections are ANSI class 300 NPS 4 and ANSI class 800 (600) NPS 3/4.

## 4.4 Test Procedure

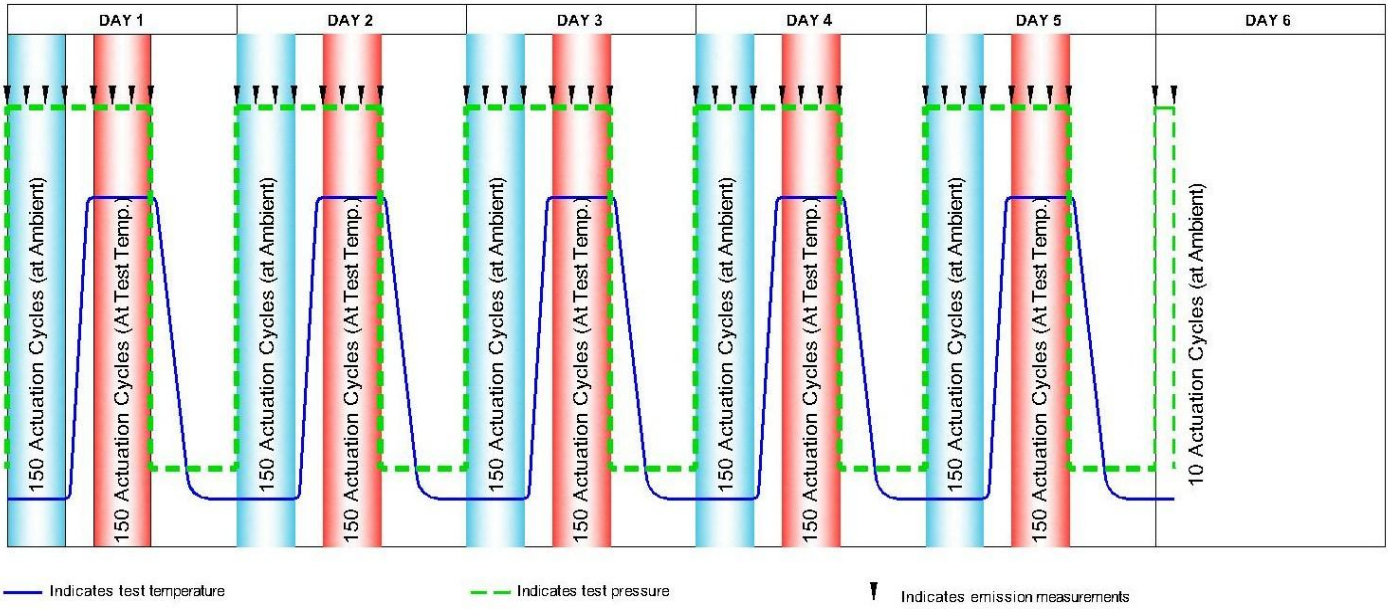
### 4.4.1 Test Fluid

#### 4.4.1.1 High temperature test

The test fluid used shall be dry methane gas, 97% minimum purity, subjected to a temperature range from ambient to 260ºC (500ºF) and pressures from 0 to 4,137 kPag (0 to 600 psig).

#### 4.4.1.2 Low temperature test

The test fluid used shall be dry methane gas, 97% minimum purity, subjected to a temperature range from ambient to 177ºC (350ºF) and pressures from 0 to 4,137 kPag (0 to 600 psig).




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Figure 2

### 4.4.2 Mechanical and Thermal Cycling

**4.4.2.1** Packing in test rigs and test valves shall be subject to a total of 1510 mechanical cycles and 5 thermal cycles per Figure 2. Mechanical and thermal cycling shall begin with the test fixture or the test valve at ambient temperature. The testing will be performed as described below.

#### 4.4.2.2 Test Profile

a) Test duration of 5 days.

b) 300 mechanical cycles per day.

c) 1 temperature cycle per day.

d) High Temperature Test:

- Pressure stays at 4,137 kPag ±34kPa (600 psig ±5 psig)
- 150 cycles at ambient
- 150 cycles at 260ºC ±3ºC (500°F ±5ºF)

e) Low Temperature Test:

- Pressure stays at 4,137 kPag ±34kPa (600 psig ±5 psig)
- 150 cycles at ambient
- 150 cycles at 177ºC ±3ºC (350°F ±5ºF)

f) Reduce the pressure to zero at the end of each day (thermal cycle).

g) Minimum daily cycling is to be divided up as follows:

- 3 hours for ambient cycling at pressure.
- Max. 2 hours to bring up to temperature (no cycling during this time).
- 3 hours for at high temperature cycling.
- Daytime or Overnight cool down with no cycling during this period

Note: This test procedure requires a maximum of 8 hours of monitoring per day.

### 4.4.3 Leak Measurement

**4.4.3.1** Leak measurements shall be conducted initially at the start of each day and at the completion of every 50 cycles. A minimum of ten (10) readings shall be taken over a one (1) minute duration. The average reading shall be

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calculated and compared with the allowable leakage. For the stem static leakage measurement, if any reading is more than fifty percent (50%) greater than the average, the readings shall be repeated.

**4.4.3.2** For the packing test in the test fixture, the leak measurement shall be conducted using a fixed detection probe located at the 12 o'clock position directly above the potential leak point as per Figure 3.

Note: If a valve test is performed to verify packing performance, the leak measurement shall be performed in compliance with EPA Method 21.

**4.4.3.3** Leak measurements shall be taken while the stem is in the static and dynamic condition. Dynamic condition measurements shall be taken on the opening and closing cycles and the average of both shall be recorded.

**4.4.3.4** For the packing test in the test fixture, the connection to the fixture and methane leak detector shall be made using tubing having the same inside diameter as the standard leak probe and connected as shown in Figures 3 and 4.

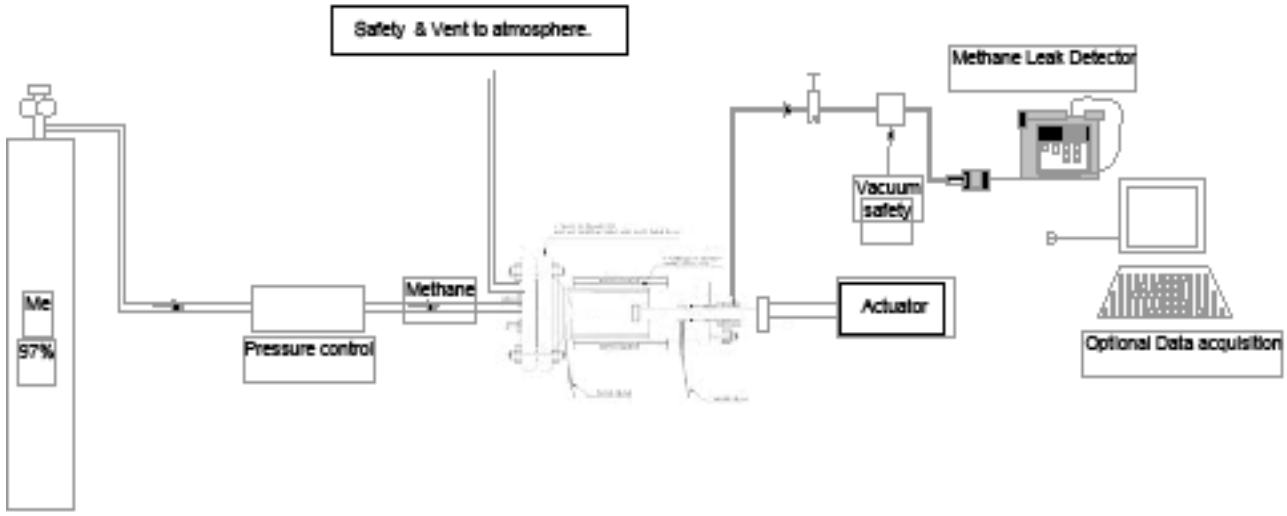



Figure 3

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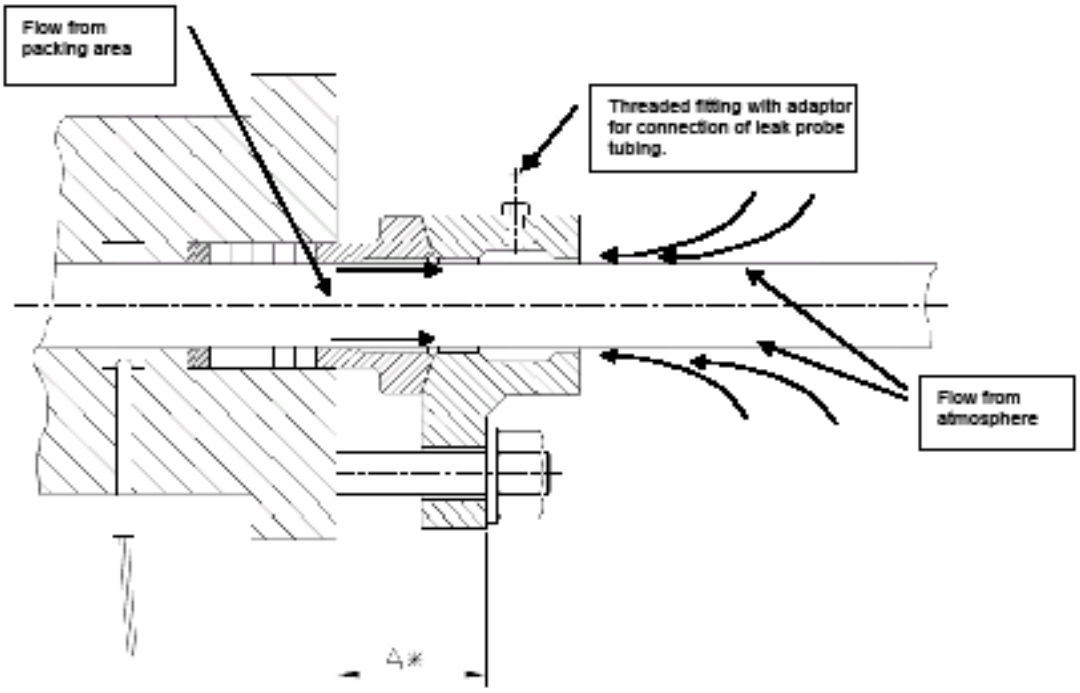



Figure 4

**4.4.3.5** The emissions leak test system shall conform to all local and governmental safety standards and shall be equipped with pressure relief valve(s) / rupture disc and vents.

### 4.4.4 Packing Adjustment

**4.4.4.1** The packing assembly must be re-adjusted to the original manufacturer's suggested gland load if the leakage rate exceeds 100 PPMV based on the averaged static condition at each 50 cycle reading. Other than the initial packing gland loading prior to starting the test, adjustments shall not be made until this criterion is met.

a) Mechanical cycling shall be discontinued during a necessary adjustment.

b) The gland flange height shall be measured and recorded before and after gland adjustment is made.

c) Continue test.

d) No additional packing adjustments are permitted. If the leak rate exceeds 100 PPMV again, based on the averaged static condition at each following 50 cycle reading, the test is over.

**4.4.4.2** Adjustments shall be reported as packing gland bolt torque values and as measured by recording the change in clearance between the gland flange and the top of the bonnet (reference dimension A* per Figures 1 and 4) or by a fixed height gauge. The gauge shall provide readout in 0.025 mm (0.001 in.) increments.

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**4.4.4.3** Flats of adjustment shall be indicated by use of a line marked on the gland flange and gland nuts at the start of the test.

### 4.4.5 Recording and Documentation

**4.4.5.1** Fugitive emissions test results shall be provided on the Fugitive Emissions Test Report Summary provided in Appendix A.

a) Leak measurements, static and dynamic, shall be recorded at the beginning of the test and at established intervals throughout the test, as required per Figure 2.

b) The packing adjustment shall be recorded and the cycle number noted along with the gland bolt torque prior to the adjustment. The gland bolt torque is for information only.

## 5 Corrosion Test

### 5.1 CORROSION TEST OVERVIEW

The corrosion test provides methods for evaluation of "cold" and "hot" corrosion caused by moisture in the packing. It also provides a means for evaluating the effect of inhibitor systems / valve stem metallurgy combinations with respect to corrosion rate and weight loss.

### 5.2 PRE-TEST REQUIREMENTS

Packing sets submitted for corrosion testing shall have an accompanying materials analysis providing details for each style of packing contained in the set. Details shall include:

a) Primary material used in manufacture

b) Type of corrosion inhibitor(s)

c) Inhibitor content by weight

d) Method of application/distribution of inhibitors in/on the packing (i.e. active or passive)

e) Packing test specimen size shall be identical to that required for leak testing.

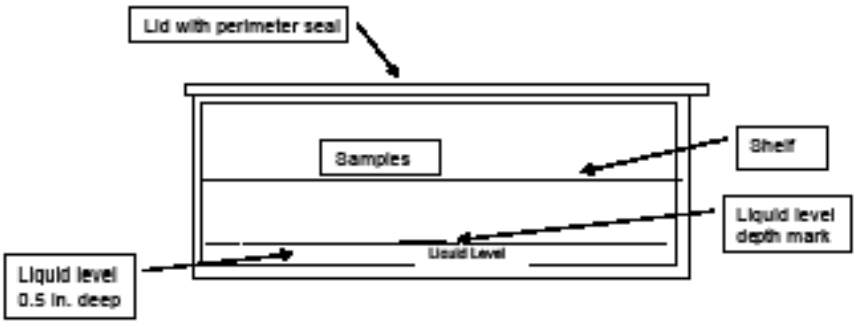



Figure 5

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### 5.3 AMBIENT CORROSION TESTING

**5.3.1** The ambient corrosion testing vessel shall consist of an enclosure large enough for containment of several sample fixtures. The enclosure shall be equipped with a shelf or rack suspended above a liquid bath as shown in Figure 5.

**5.3.2** The compression fixture shall be as shown in Figure 6.

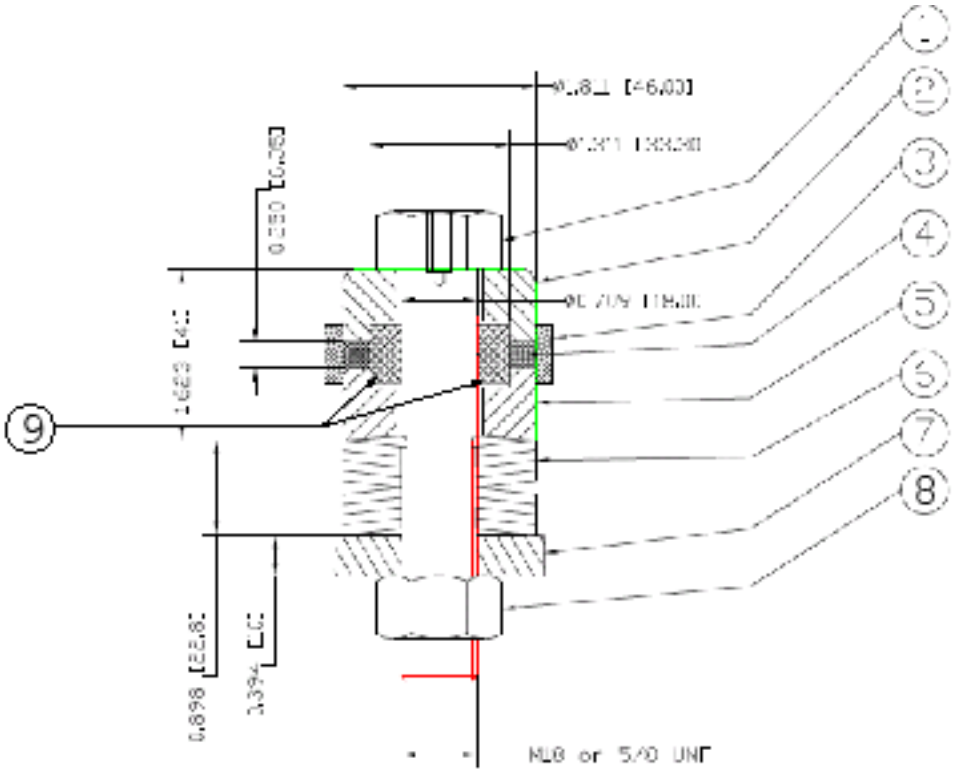



Figure 6

**5.3.2**

**5.3.2.1** Compression test fixture components

1) Bolt
2) Top housing
3) Packing retainer
4) Packing
5) Lower housing
6) Disc springs

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7) Washer

8) Nut

9) Metal test specimen

**5.3.3** Prior to the start of corrosion testing, the compression fixture shall be:

a) Cleaned in an ultrasonic acetone bath

b) Adjusted so as to provide a 30 MPa (4350 psi) ±0.69 Mpa (100 psi) compressive stress on the test packing.

**5.3.4** Prior to assembly of test set, packing samples shall be wetted by soaking in de-mineralized water for 24 hours, creating a damp environment at ambient temperature, 22ºC ±11ºC (72ºF ±20ºF).

**5.3.5** The test vessel shall be filled to a level of 1.27 cm (0.5 in).

**5.3.6** Test samples shall be assembled on the compression fixtures and placed into the ambient corrosion test vessel.

**5.3.7** The test packing shall be installed around a test specimen (metal ring), representing the valve stem material being evaluated.

a) Sample steel rings shall be machined from metal rods having the same properties as the finished valve stem. A common material selection is 410 stainless steel (13% Chrome).

b) Nominal dimensions of the machined sample shall be according to MSS SP-120.

c) Nominal finish shall be 0.4 – 0.8 µm $R_a$ (16 – 32 µ-in. $R_a$).

**5.3.8** The duration of the test shall be 28 days.

## 5.4 HIGH TEMPERATURE CORROSION TESTING

**5.4.1** The high temperature corrosion test rig shall provide a heated chamber for multiple individual test fixtures containing samples of metal and packing. See Annex 2.

**5.4.2** The housing shall be equipped with electric heating elements or a heater blanket that surrounds the outer periphery of the device, and allows for insertion of multiple test fixtures. See Figure 7 and Annex 2.

**5.4.3** Each test fixture shall be equipped with a vapor feed line, providing constant replenishment of fluid to the test samples.

**5.4.4** Prior to the start of corrosion testing, the compression fixture shall be adjusted so as to provide a 30 Mpa (4350 psi) compressive stress on the test packing.

**5.4.5** Packing samples shall be subjected to de-mineralized water at a test temperature of 149ºC ±17ºC (300ºF, ±30ºF). The water pressure shall be maintained at 45 barg ±2.25 bar (650 psig ±32.5 psi).

**5.4.6** The duration of the test shall be 35 days.

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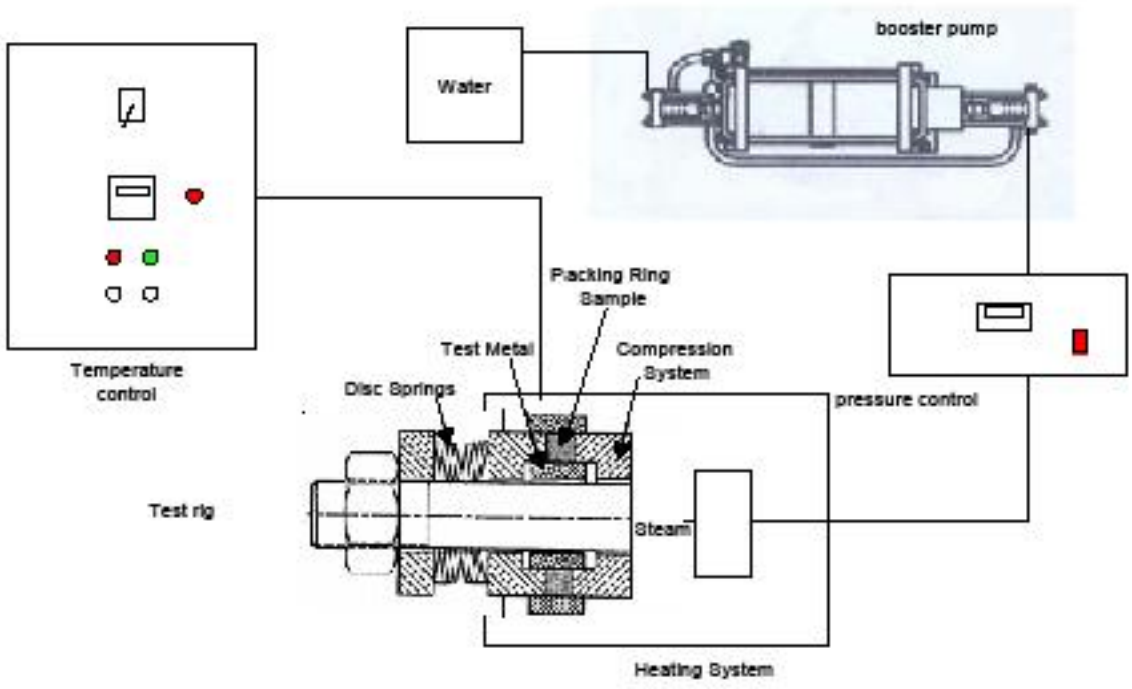



Figure 7—High Temperature Corrosion Test System

**5.4.6**

## 5.5 CORROSION TEST REPORTING

Corrosion test data shall be reported on the Corrosion Test Data Sheet provided in Appendix A. The report shall include the following information:

a) Photographic record of each sample at 100X and 200X magnification.

b) Descriptive report on the estimated degree of stem/shaft pitting in ten percent (10%) increments of the surface area, which shall include the mean pit depth and the maximum pit depth.

c) Descriptive report on the estimated degree of adhesion in ten percent (10%) increments of surface area.

# 6 Packing Materials Test

The packing materials test considers weight loss, density, lubricant content, and leachables as per the procedures outlined below. For new configurations, test packing shall be selected at random from a minimum production lot as supplied by the manufacturer. For existing configurations, test packing shall be provided by a blind selection of production product from the manufacturer's normal distribution network. Validation of the random or the blind selection of process shall be provided to the testing lab. Random selection does not apply to testing of prototype material.

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## 6.1 WEIGHT LOSS

### 6.1.1 GRAPHITE FOIL

a) Conducted in a controlled environment using Thermo Gravimetric Analysis (TGA) tester.

b) Flexible graphite sample size shall be between 0.5 grams and 3.5 grams.

c) Three samples shall be tested at same time. Record weight of each sample.

d) Samples shall be preconditioned for one hour at 150ºC (302ºF) with a ramp up speed of 10ºC (18ºF) per minute.

e) After one hour, the samples shall continue ramping up at 10ºC (18ºF) per minute to the final test temperature of 593ºC (1100ºF).

f) This test temperature shall be held for 24 hours and then cooled.

g) Weigh samples after cooling and record weight. TGA testers can weigh samples without being removed from the heat source which is acceptable.

h) Determine the percent weight loss of each sample and record.

i) Average the results and a weight loss greater than 15% is not acceptable.

Note: This test method also follows an established testing standard, FSA-G-604-07, issued by the Fluid Sealing Association.

### 6.1.2 BRAIDED PACKING RING

a) Conducted in an oven with full exposure to air. (Oxygen-rich environment.)

b) Select a test ring of a sample packing set. If the packing set is comprised of more than one type of packing ring, each type shall be tested.

c) Record weight.

d) Preheat oven to 150ºC (300ºF).

e) Place sample in oven for one hour.

f) Remove samples, cool to room temperature, and record weight.

g) Increase oven temperature to 260ºC (500ºF) and repeat steps e and f.

h) Increase temperature to 538ºC in 56ºC (1000ºF in 100ºF) increments, repeating steps e and f after each increase.

i) Discontinue test if weight loss exceeds 50%.

## 6.2 DENSITY

Packing Density shall be determined by dividing the sample weight by the sample volume, with the sample volume determined as follows:

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a) Braided packing volume shall be determined by measuring the sample length, width and thickness.

$$\text{Volume} = \text{Length} \times \text{Width} \times \text{Thickness}$$

b) Die-formed packing (ring form) volume shall be determined by the following equation:

$$\text{Volume} = [(OD^2 - ID^2) \times \text{Thickness} \times p] / 4$$

## 6.3 LUBRICANT CONTENT

**6.3.1** PTFE Content:

PTFE content shall be established by determining the % total fluorine in the packing, and comparing with a base fluorine percentage of 76, as follows:

a) Determine total percent of fluorine content using ASTM D1179 or ASTM D4327.

b) Divide total percent of fluorine as obtained in (a) above by 0.76 to obtain the approximate percent of PTFE content.

**6.3.2** Wet Lubricant:

Wet lubricant percentage shall be determined by dividing the "weight of the lubricant extract" by the "original sample weight" and multiplying by 100. The procedure shall be as follows:

a) Cut samples into short lengths, and record the total sample weight. (Approx. 15 grams is required)

b) Place cut samples into an extraction thimble and insert thimble into Soxhlet extraction unit that is nearly filled with a solvent such as methylene chloride.

c) Turn on water flow through the condenser and bring the methylene chloride to a moderate boil through the use of the "low" setting on a hot plate. Extract for approximately 16 hours.

d) Record the weight of a dry evaporation dish.

e) Pour the Methylene Chloride extract solution into the dry evaporating dish and evaporate over a steam bath. Continue to pour the extract solution from the flasks into the dish until all the methylene chloride has been evaporated.

f) Place the evaporation dish in a hot-air oven set between 100°C – 121°C (212°F – 250°F) for 30 minutes. Cool the evaporation dish to room temperature in a desiccator.

g) Record the weight of the evaporation dish and lubricant extract.

h) Calculate the lubricant weight by subtracting the weight of the dry evaporation dish per (step d) from the combined weight in step (g).

Note: A reference test method for Soxhlet Extraction can be found in ASTM C613.

## 6.4 LEACHABLES

Leachables testing shall be per the following:

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a) Packing submitted for testing shall contain a corrosion inhibitor.

b) Chloride Testing shall be per ASTM D512

c) Leachable Fluoride Testing shall be per ASTM D1179 or ASTM D4327

d) Proof of testing and analysis shall be maintained and provided with the final test report.

### 6.5 PACKING MATERIALS TEST REPORTING

Packing Materials Test data shall be reported on the Material Test Data Sheet per Appendix A.

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# APPENDIX A—TEST DOCUMENTATION

## A.1 FUGITIVE EMISSIONS TEST REPORT SUMMARY

## A.2 AMBIENT TEMPERATURE CORROSION TEST DATA SHEET

## A.3 HIGH TEMPERATURE CORROSION TEST DATA SHEET

## A.4 MATERIALS TEST REPORT

A.1—Fugitive Emissions Test Report Summary

<table>
<tr><td colspan="3">API Std 622<br><br>Fugitive Emissions Testing Report Number:_____</td></tr>
<tr><td colspan="2">Application Profile: Check One<br><br>o Rotating o Rising</td><td>Manufacturer:<br><br>Description:</td></tr>
<tr><td colspan="2">Testing Facility:<br><br>Technician:<br><br>Witness:<br><br>Start Date: Completion:</td><td>Source: Date:<br><br>o Manufacturer<br><br>o Distributor<br><br>Packaged:</td></tr>
<tr><td>Gland Load<br><br>Information<br>psi.<br><br>kPa</td><td>Gland Nut Torque:<br><br>ft-lbs<br><br>in-lbs<br><br>Nm</td><td>Indicate New or Current Product<br><br>o New<br><br>o Current</td></tr>
<tr><td colspan="3">Notes concerning installation instructions</td></tr>
</table>

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| Testing Profile Details | | | | | | |
|---|---|---|---|---|---|---|
| API 622 | Testing Data | For Test Rig | For Test Valve | | | |
| Test Segment | Leak Measurement Static/Dynamic | Temperature | Reference Temperature at Packing Gland | Flats Adjusted–Gland Nut Torque ft-lbs & Reference A Height | | |
| Day 1 Start, Ambient 0 – 150 cycles P = _____ | _____ | _____ | _____ | _____ | _____ | _____ |
| | _____ | _____ | _____ | _____ | _____ | _____ |
| | _____ | _____ | _____ | _____ | _____ | _____ |
| | _____ | _____ | _____ | _____ | _____ | _____ |
| | _____ | _____ | _____ | _____ | _____ | _____ |
| Low or High Temperature 151 – 300 cycles P = _____ | _____ | _____ | _____ | _____ | _____ | _____ |
| | _____ | _____ | _____ | _____ | _____ | _____ |
| | _____ | _____ | _____ | _____ | _____ | _____ |
| | _____ | _____ | _____ | _____ | _____ | _____ |
| | _____ | _____ | _____ | _____ | _____ | _____ |
| Day 2 Start, Ambient 301 – 450 cycles P = _____ | _____ | _____ | _____ | _____ | _____ | _____ |
| | _____ | _____ | _____ | _____ | _____ | _____ |
| | _____ | _____ | _____ | _____ | _____ | _____ |
| | _____ | _____ | _____ | _____ | _____ | _____ |
| | _____ | _____ | _____ | _____ | _____ | _____ |
| Low or High Temperature 451 –600 cycles P = _____ | _____ | _____ | _____ | _____ | _____ | _____ |
| | _____ | _____ | _____ | _____ | _____ | _____ |
| | _____ | _____ | _____ | _____ | _____ | _____ |

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| | | | | | | |
|---|---|---|---|---|---|---|
| | _____ | _____ | _____ | _____ | _____ | _____ |
| | _____ | _____ | _____ | _____ | _____ | _____ |
| Day 3 Start, Ambient<br><br>601 - 750 cycles<br>P = _____ | _____ | _____ | _____ | _____ | _____ | _____ |
| | _____ | _____ | _____ | _____ | _____ | _____ |
| | _____ | _____ | _____ | _____ | _____ | _____ |
| | _____ | _____ | _____ | _____ | _____ | _____ |
| | _____ | _____ | _____ | _____ | _____ | _____ |
| | _____ | _____ | _____ | _____ | _____ | _____ |
| Low or High Temperature 751 - 900 cycles<br>P = _____ | _____ | _____ | _____ | _____ | _____ | _____ |
| Day 4 Start, Ambient<br><br>901 - 1050 cycles<br>P = _____ | _____ | _____ | _____ | _____ | _____ | _____ |
| | _____ | _____ | _____ | _____ | _____ | _____ |
| | _____ | _____ | _____ | _____ | _____ | _____ |
| | _____ | _____ | _____ | _____ | _____ | _____ |
| | _____ | _____ | _____ | _____ | _____ | _____ |
| | _____ | _____ | _____ | _____ | _____ | _____ |
| Low or High Temperature 1051 - 1200 cycles<br>P = _____ | | _____ | _____ | _____ | _____ | _____ |

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| | | | | | | |
|---|---|---|---|---|---|---|
| | _____ | | | | | |
| Day 5 Start, Ambient 1201 - 1350 cycles P = _____ | _____ | _____ | _____ | _____ | _____ | _____ |
| | _____ | _____ | _____ | _____ | _____ | _____ |
| | _____ | _____ | _____ | _____ | _____ | _____ |
| | _____ | _____ | _____ | _____ | _____ | _____ |
| | _____ | _____ | _____ | _____ | _____ | _____ |
| | _____ | _____ | _____ | _____ | _____ | _____ |
| Low or High Temperature 1351 - 1500 cycles P = _____ | _____ | _____ | _____ | _____ | _____ | _____ |
| Day 6 Start, Ambient 1501 - 1510 cycles P = _____ | _____ | _____ | _____ | _____ | _____ | _____ |
| | _____ | _____ | _____ | _____ | _____ | _____ |
| | _____ | _____ | _____ | _____ | _____ | _____ |
| | _____ | _____ | _____ | _____ | _____ | _____ |
| | _____ | _____ | _____ | _____ | _____ | _____ |
| | _____ | _____ | _____ | _____ | _____ | _____ |

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<table>
<tr><td colspan="2">API Std 622<br><br>Emissions Testing Report Summary</td></tr>
<tr><td>Test Number:</td><td>Test Date:</td></tr>
<tr><td>Packing Material:</td><td>Style Number:</td></tr>
<tr><td>Packing Manufacturer:</td><td>Source of Sample:</td></tr>
<tr><td rowspan="2">Test Packing Cross-section:</td><td>Laboratory Name:</td></tr>
<tr><td>Location of Test:</td></tr>
<tr><td>Packing Gland OD and ID (at the packing):<br><br>OD = ID =</td><td>Packing Gland Bolt Diameter =<br><br>Packing Material:</td></tr>
<tr><td rowspan="2">Number of Mechanical Cycles:<br><br>Gland Load:</td><td>Packing Compression % of Free Height =</td></tr>
<tr><td>Torque on Gland Nuts (each side) = _____/_____ as installed</td></tr>
<tr><td>Number of Thermal Cycles:</td><td rowspan="2">Mechanical Cycles Prior to Readjustment:<br><br>Gland Nut Torque before readjustment</td></tr>
<tr><td>Maximum Test Pressure:</td></tr>
</table>

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| | |
|---|---|
| | |
| Packing Configuration:<br><br>Number of rings tested:<br>Circle the following<br>Ring shape (square, circular, vee)<br>Solid or split<br>Braided<br>Die formed<br>Spool stock<br><br>Wire or other reinforcement<br><br>Corrosion inhibitor & type<br><br>Other | Show Sketch of Packing Installation-define each ring: |

A.2—Ambient Temperature Corrosion Test Data Sheet

| | | |
|---|---|---|
| API Std 622<br><br>Ambient Corrosion Testing: Materials Qualification and Functional Testing Report | | |
| Product Manufacturer: | Materials Analysis:<br><br>Laboratory: _____<br><br>Test Number: _____ | Date: |
| Pre-Qualification Analysis Information | | |
| Primary Material of Manufacture: | Corrosion Inhibitor(s) | Inhibitor Volume by Weight |

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| Corrosion Testing Facility | Test Technician | Date<br>Start/Complete |
|---|---|---|
| Ambient Corrosion Testing | Load Stress: | S_____ C_____ |
| Metal Sample Description | Observations | |
| 1. | | S_____ C_____ |
| 2. | | S_____ C_____ |
| 3. | | S_____ C_____ |
| 4. | | S_____ C_____ |
| 5. | | S_____ C_____ |
| 6. | | S_____ C_____ |
| 7. | | S_____ C_____ |
| 8. | | S_____ C_____ |
| All reports shall include the following:<br>1. Microscopic view photographic record of results. Magnification levels of 100X and 200X.<br>2. X-ray corrosion analysis (in graphic form.)<br>3. Descriptive report on the degree of stem/shaft pitting that occurred<br>a. Percent of surface area that is corroded<br>b. The mean and maximum pit depth | | |
| A schematic of the actual test vessel shall be provided.<br>For tests having greater numbers of samples than provided for on this form, the testing lab will fill in a separate form for each sample tested. | | |

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## A.3—High Temperature Corrosion Test Data Sheet

| | | |
|---|---|---|
| API Std 622<br><br>High Temperature Corrosion Testing: Materials Qualification and Functioning Testing Report | | |
| Product Manufacturer: | Materials Analysis:<br><br>Laboratory:<br><br>Test Number: _____ | Date: |
| | Pre-Qualification Analysis Information | |
| Primary Material of Manufacture: | Corrosion Inhibitor(s) | Inhibitor Volume by Weight |
| Corrosion Testing Facility | Test Technician | Date<br>Start/Complete |
| | Load Stress: | S_____ C_____ |
| Metal Sample Description | Obeservations | |
| 1. | | S_____ C_____ |
| 2. | | S_____ C_____ |
| 3. | | S_____ C_____ |
| 4. | | S_____ C_____ |

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| | | |
|---|---|---|
| 5. | | S_____ C_____ |
| 6. | | S_____ C_____ |
| 7. | | S_____ C_____ |
| 8. | | S_____ C_____ |

All reports shall include the following:
1. Microscopic view photographic record of results. Magnification levels of 100X and 200X.
2. X-ray corrosion analysis (in graphic form.)
3. Descriptive report on the degree of stem/shaft pitting that occurred
   a. Percent of surface area that is corroded
   b. The mean and maximum pit depth

A schematic of the actual test vessel shall be provided.
For tests having greater numbers of samples than provided for on this form, the testing lab will fill in a separate form for each sample tested.

A.4—Materials Test Report

API Std 622

Graphite Foil Test Report

| Testing Laboratory | | Technician | Date |
|---|---|---|---|
| Product Name/Model | Base Composition of Material<br><br>Graphite Purity<br><br>Ash Content<br><br>Sulphur<br><br>Chloride | Source:<br><br>o Manufacturer<br><br>o Distributor | Date: |

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| | | | |
|---|---|---|---|
| | Fluoride<br><br>Corrosion Inhibitor<br><br>Oxidation Inhibitor | <br><br>Nominal Thickness<br><br>Density | |
| Indicate New or Current Product | o New<br><br>o Current | | |
| Weight Loss (TGA) Testing | | | |
| Measurement Equipment Name and Manufacturer(s) | | | |
| Calibration Dates on TGA: | | | |
| Start Sample Weight: Sample #1 | Sample weight | | |
| Weight after 1 hour preconditioning at 150°C (302°F) | Sample Weight<br>Variation, weight _____% of initial weight_____ | | |
| Weight after 24 hours at 593°C (1100°F) | Sample Weight<br>Variation, weight _____% of initial weight | | |
| Start Sample Weight: Sample #2 | Sample weight | | |
| Weight after 1 hour preconditioning at 150°C (302°F) | Sample Weight<br>Variation, weight _____% of initial weight_____ | | |
| Weight after 24 hours at 593°C (1100°F) | Sample Weight<br>Variation, weight _____% of initial weight | | |

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| | |
|---|---|
| Start Sample Weight: Sample #3 | Sample weight |
| Weight after 1 hour preconditioning at 150°C (302°F) | Sample Weight<br>Variation, weight _____% of initial weight_____ |
| Weight after 24 hours at 593°C (1100°F) | Sample Weight<br>Variation, weight _____% of initial weight |
| Average weight loss of three samples | Variation, weight _____% of initial weight |

| | | | |
|---|---|---|---|
| API Std 622<br><br>Packing Materials Test Report<br>(separate test report required for different packing ring requirements) | | | |
| Testing Laboratory | | Technician | Date |
| Product Name/Model | Base Composition of Material | Source:<br><br>o Manufacturer<br><br>o Distributor<br><br>Packaged: | Date: |
| Indicate New or Current Product | o New<br><br>o Current | Description of Product: o Die-formed o Braided o Other | |

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| Weight Loss (Oven) Testing | |
|---|---|
| Measurement Equipment Name and Manufacturer(s) | |
| Calibration Dates on Oven Recorders:<br><br>Calibration Dates on Weighing Equipment: | |
| Start Sample Weight: | |
| Weight after one hour at 149ºC (300°F). | Variation, weight _____%_____ |
| Weight after one hour at 260ºC (500°F). | Variation, weight _____%_____ |
| Weight after one hour at 315ºC (600°F). | Variation, weight _____%_____ |
| Weight after one hour at 371ºC (700°F). | Variation, weight _____%_____ |
| Weight after one hour at 427ºC (800°F). | Variation, weight _____%_____ |
| Weight after one hour at 482ºC (900°F). | Variation, weight _____%_____ |
| Weight after one hour at 538ºC (1000°F). | Variation, weight _____%_____ |

| API Std 622<br><br>Packing Density | |
|---|---|
| Calibration Date: | Braided Construction Volume<br>$V_{bc}$ = L x W x T |
| Sample Weight: | Die Formed Packing Ring Volume:<br>$V_{df}$ = [(OD$^2$ – ID$^2$) x Thickness x p] / 4 |

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<table>
<tr><td colspan="4">Density = Sample Weight / Sample Volume =</td></tr>
<tr><td colspan="4">Lubricant Content</td></tr>
<tr><td colspan="2">Packing Lubricant Content</td><td colspan="2">Volume %</td></tr>
<tr><td colspan="4">Leachables</td></tr>
<tr><td>ASTM D 512 Chlorides</td><td>Value:</td><td>Lab Name:</td><td>Date:</td></tr>
<tr><td>ASTM D 1179 or<br>ASTM D 4327 Fluorides</td><td>Value:</td><td>Lab Name:</td><td>Date:</td></tr>
<tr><td colspan="2">Corrosion Inhibitor</td><td colspan="2">Type(s):</td></tr>
<tr><td colspan="4">Certification of Selected Process:</td></tr>
<tr><td colspan="2">New Configurations:</td><td colspan="2">Existing Configurations:</td></tr>
<tr><td colspan="2">Validation of Random Selection:<br><br>Y_____ N_____<br><br>Lot Size:<br><br>Signature:</td><td colspan="2">Validation of Random Selection<br><br>Distributor:<br><br>Signature:</td></tr>
</table>

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# ANNEX 1

## 1.0 EMISSIONS TEST FIXTURE CONSTRUCTION

**1.1** The test fixture shall be constructed to simulate a Class 300 DN 100 (NPS4) block valve.

**1.2** Flanges shall be carbon steel, Class 300 DN 100 (NPS4) per ASME B16.5 Table F2-1.1 minimum.

**2.4** Gaskets shall be spiral wound type, DN 100 (4 in.) Class 300 per ASME B16.20 with flexible graphite.

**1.4** The body of the test fixture shall be made from NPS4 carbon steel seamless pipe having schedule 80 minimum pipe schedule or from A-105 or other suitable carbon steel materials.

**1.5** The body of the assembly shall provide sufficient clearance to enable movement of the shaft, of 0.19 m (7.62 in.), minimum.

**1.6** The packing gland shall be ASTM A216 Grade WCB or from A-105 or other suitable carbon steel materials.

**1.7** Stem shall be ASTM A182 Grade F6a Rc15-28.8 (200-275 HB).

**1.8** Gland bolts/studs: ASTM A193 Grade B7.

**1.9** Gland nuts: ASTM A194 Grade 2H.

**1.10** Heating shall be provided by external heating blankets or coils.

**1.11** Insulation may be applied to the body, flanges and gland areas.

**1.12** A primary measurement thermocouple shall be positioned to monitor temperature of the fixture body.

**1.13** A reference measurement thermocouple shall be positioned as dimensioned in Figure A-1.

**1.14** The emissions measuring equipment shall be coupled to the modified gland (flange) using flexible tubing with an identical inside diameter as that of the detection probe tubing supplied with the detector.

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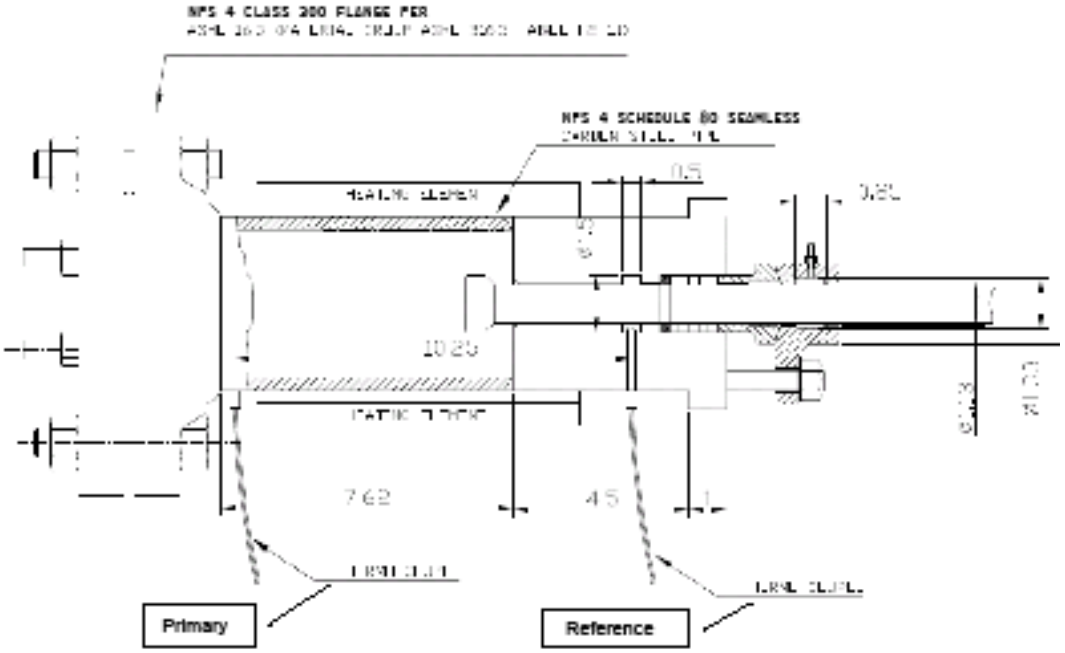



Figure A-1

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# ANNEX 2

## 2.0 High Temperature Corrosion Test Fixture Construction

**2.1** The fixture may be constructed to allow for multiple test assemblies.

**2.2** The fixture design shall be designed to accommodate a single ring of packing having a 6.35 mm (0.250 in.) square cross-section.

**2.3** The fixture housing shall be fabricated from ASTM A216 Grade WCB.

**2.4** The fixture bolt(s) shall be ASTM A182 Grade F6a Rc15-28.8 (200-275 HB).

**2.5** Fixture nut(s) shall be ASTM A194 Grade 2H.

**2.6** Compression gland load nut shall be ASTM A216 Grade WCB.

**2.7** Compression gland top piece shall be ASTM A216 Grade WCB.

**2.8** A buffer seal washer shall be located between the housing and pressurization housing.

**2.9** The fixture shall be equipped with a fluid flow transition fitting which shall be fabricated from ASTM A216 Grade WCB.

**2.10** Belleville washers shall be used to maintain a compressive load of 30 Mpa (4350 psi).

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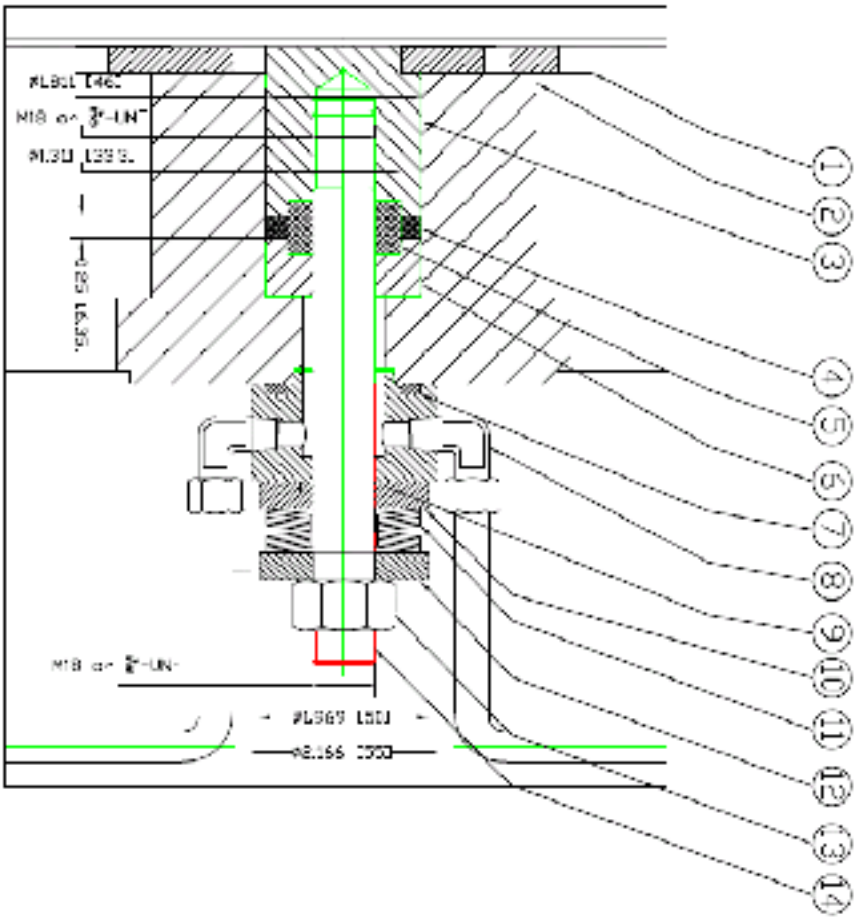



Figure A-2

**2.10**

**2.11** Components (see Figure A-2)

1. Housing retainer
2. Fixture housing
3. Compression gland bottom piece
4. Packing sample
5. Metal test sample
6. Compression gland top piece
7. Seal

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8. Tubing and fittings
9. Seal
10. Washer
11. Disc springs
12. Washer
13. Nut
14. Bolt / shaft

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